

pjax-pdf

Markdown → PDF with Mermaid, math and Tailwind

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Text and inline formatting

Standard Markdown works as expected: **bold**, *italic*, `inline code`, ~~strikethrough~~, and links.

Typographic replacement turns “quotes” into curly ones and – into en dashes.

Blockquotes, like every other element, are styled against the `.doc` wrapper in `theme.css` rather than through `@tailwindcss/typography`'s `prose` class.

Tailwind classes in Markdown

Because Tailwind scans the generated HTML, utility classes work three ways.

Via markdown-it-attrs

That heading carries `{.text-blue-600}` in the source.

Via inline HTML

Note the `break-inside-avoid` class — Tailwind's page-break utilities work here, so this card row is kept whole rather than split across pages.

9,836

UNITS ON HAND

1,618

WRITTEN OFF

2,440

NET CHANGE

Via the document wrapper

Classes on the `.doc` wrapper in `wrapDocument()` apply to the whole document; edit `theme.css` to change the house style for every document at once.

Syntax highlighting

Handled by `highlight.js` at build time, so there is no browser work and the output is deterministic.

```
export async function renderPdf(html: string, out: string): Promise<void> {
  const browser = await puppeteer.launch({ args: ['--no-sandbox'] });
  const page = await browser.newPage();
  await page.setContent(html, { waitUntil: 'networkidle0' });
  await page.pdf({ path: out, format: 'A4', printBackground: true });
  await browser.close();
}
```

```
select sku, sum(quantity) as qty
from inventory_movement
where warehouse = 'north'
group by sku
order by qty desc;
```

Mathematical notation

TeX is typeset by **KaTeX** at build time, alongside the syntax highlighter — the HTML that reaches the browser already contains the finished formulas.

Inline math goes between single dollars: the harmonic mean of P and R is $F = \frac{2PR}{P+R}$, and a point $x \in \mathbb{R}^n$ has $\|x\|_2 = \sqrt{\sum_i x_i^2}$.

Display math goes between double dollars, on their own lines:

$$d^2(x, u) = (x_1 - u_1)^2 + (x_2 - u_2)^2 + \cdots + (x_n - u_n)^2$$

A fenced block tagged `math` is the same thing, for editors that prefer a fence:

$$P(c \mid x) \propto P(c) \prod_{j=1}^m P(x_j \mid c)$$

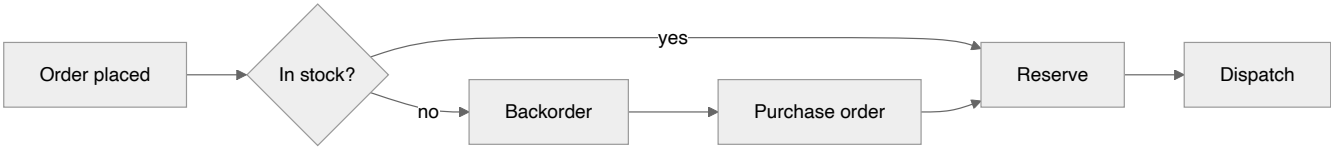
Environments, alignment and boxed results all work:

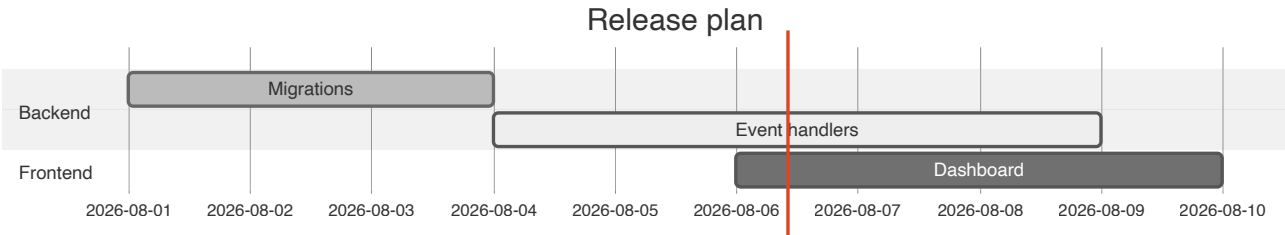
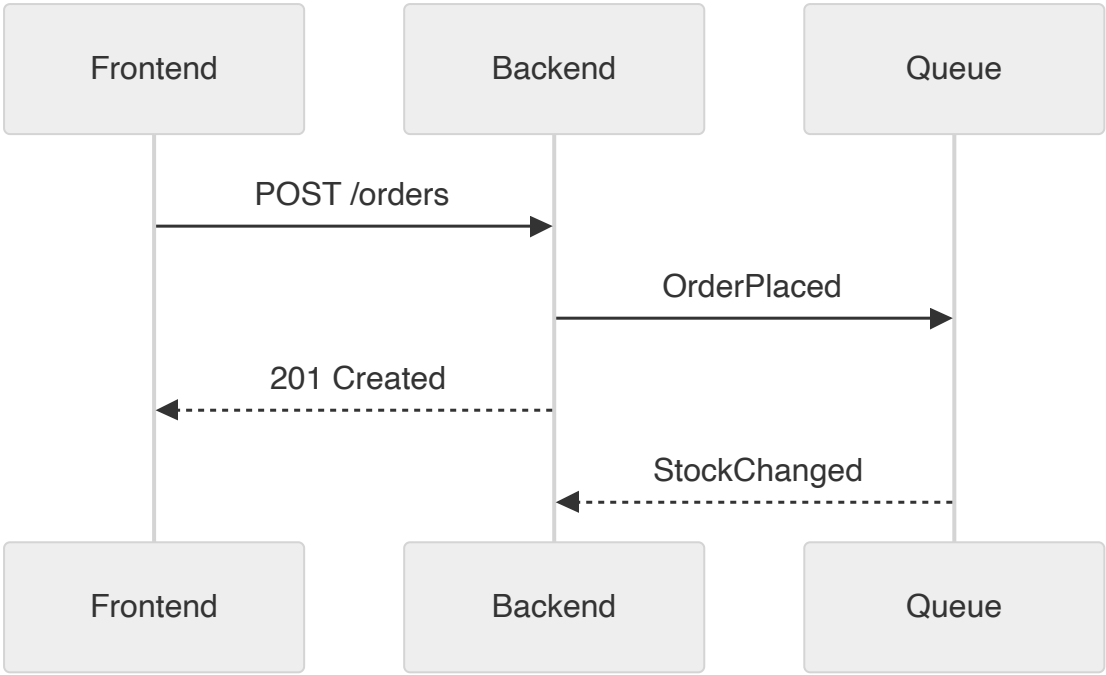
$$\begin{aligned} \text{acc} &= \frac{TP + TN}{TP + FP + FN + TN} \\ P &= \frac{TP}{TP + FP} \qquad R = \frac{TP}{TP + FN} \\ F &= \frac{2PR}{P + R} = \boxed{\frac{25}{33}} \approx 0.76 \\ A &= \begin{bmatrix} 3 & 2 & 1 \\ -1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \qquad \Theta(n \log n) \leq \Theta(n^2) \end{aligned}$$

To write a literal dollar sign, escape it: `$4.99` stays text.

Mermaid diagrams

Each fence is rendered in-page and **awaited** before the PDF is snapshotted — the step that `md-to-pdf` gets wrong.





Tables

Warehouse	Region	Location prefix	Units
north-1	North	N1	4,102
south-1	South	S1	6,238
south-2	South	S2	3,598

Lists

- Ordered lists survive page breaks.
- Nested items indent correctly.
 - Unordered child
 - Another child
- Task lists render as plain checkboxes:

- ☒ Mermaid rendering awaited
- ☒ Math typeset at build time
- ☒ Tailwind compiled per-document
- ☐ Cached CSS between runs